

DD_SUB-WF-ACTIVATE-01 - Subscription Activation

Developer implementation guide: DD_SUB-WF-ACTIVATE-01-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

SUB-WF-ACTIVATE-01 defines the activation operation for a subscription.

Activation moves a subscription from:

CREATED -> ACTIVE

It is the first state-changing workflow in the customer service lifecycle. After activation succeeds, the customer can receive service, billing can start, fulfillment has provisioned the network, equipment is bound, and downstream systems receive a SubscriptionActivated event.

Use this DD when implementing:

- POST /api/subscriptions/{subscription_id}/activate.
- Activation preview.
- Activation validation rules.
- subscription_activation_config.
- Activation-specific BPMN variables.
- Activation-specific Kafka events.
- Activation-specific Drools package behavior.
- The activation composition of shared workers from DD_SUB-WF-FRAMEWORK-01-WORKERS.

2. Ownership

This DD owns:

- Activation trigger taxonomy.
- Activation config table.
- Activation precondition rules.
- Cycle anchor overflow rules for activation.
- Activation backdating rules.
- Activation equipment-binding rules.
- Activation REST endpoint contract.
- Activation preview endpoint contract.
- Activation domain events: SubscriptionActivated and SubscriptionActivationFailed.
- Worker usage rules for activation BPMNs.

This DD does not own:

- Subscription master schema and state primitives. See SUB-LM-01.
- Operation orchestration tables, idempotency, concurrency, and operation status APIs. See SUB-WF-FRAMEWORK-01.
- Worker implementation contracts. See DD_SUB-WF-FRAMEWORK-01-WORKERS.
- WIK-specific BPMN sequence. See DD_SUB-WF-ACTIVATE-01-FLOW-WIK.
- Billing calculation, installation fees, activation fees, proration, tax, invoice, or wallet logic. See BIL DDs.
- Network provisioning. See FUL-03.
- HomePass claim. See FUL-07.
- Equipment instance lifecycle. See OSR-INSTANCE-01.
- Customer onboarding before subscription creation. See FUL-02.

- Notification template/channel delivery. See NOT-01.

3. Mental Model

Activation is not a simple status update.

It is a saga:

```
flowchart LR
    API[Activate endpoint] --> FW[SUB-WF Framework]
    FW --> BPMN[Activation BPMN]
    BPMN --> VALIDATE[Drools validation]
    BPMN --> LM1[SUB-LM pending status]
    BPMN --> BIL[Billing intent / payment]
    BPMN --> FUL[Fulfillment activation]
    BPMN --> HP[HomePass claim]
    BPMN --> OSR[Equipment bind]
    BPMN --> LM2[SUB-LM commit ACTIVE]
    BPMN --> EVT[Kafka event]
    BPMN --> NOTIF[Notification]
```

The subscription must not become ACTIVE until the required activation work has succeeded.

For WIK, the reference flow is pay-first:

```
validate -> PENDING_ACTIVATION -> billing/payment -> network activation -> HomePass claim -> equipment bind -> ACTIVE
```

Other operators may have different BPMNs, but they must use the same framework and worker contracts.

4. Deployment Unit Clarification

This DD does not introduce a standalone activation-service.

Activation is deployed as part of the subscription workflow capability:

- Route/controller in the SUB-WF application: POST /api/subscriptions/{subscription_id}/activate.
- Config migration/table: subscription_activation_config.
- BPMN process definition, for example sub-activate-wik.
- Drools package, for example rules.subscription.activate.wik.
- Worker usage through the shared worker catalog.

Budget deployment:

```
subscription app deployment:
- SUB-LM module
- SUB-WF framework module
- activation endpoint/controller
- activation BPMN deployment package
- worker pods/commands as defined in the WORKERS DD
```

Logical ownership still remains separate by DD, package/module, migration, route group, and metrics label.

5. Activation Triggers

Trigger	Purpose	Required role	Typical caller
BACKOFFICE_OPERATOR	Manual activation by back office.	SUBSCRIPTION_OPERATOR or SUBSCRIPTION_ADMIN	BO admin UI
CALL_CENTER	Activation requested through call center.	CALL_CENTER	Customer care UI
CUSTOMER_SELF_SERVICE	Customer starts activation from portal/app.	CUSTOMER_SELF_SERVICE	Customer portal
SALES_FIELD	Sales/installation agent activates after onboarding/install.	SALES_AGENT	FUL-02 onboarding/field flow
SCHEDULED_ACTIVATION	Scheduler activates on pre-booked date.	SCHEDULER_INTERNAL	Scheduler job

Trigger meaning:

- All triggers have the same business intent: start service.

- The trigger mainly controls audit, role-gating, and operator policy.
- Settlement differences are handled by billing intent and operator rules, not by creating separate activation operation kinds.

6. Data Model

6.1 subscription_activation_config

What this table is for:

subscription_activation_config stores operator-level activation policy. The activation endpoint, Drools rules, and BPMN use this row to decide which activation paths are allowed.

Use this table when:

- Checking whether customer self-service activation is enabled.
- Checking whether sales-field activation is enabled.
- Checking whether scheduled activation is enabled.
- Resolving the default cycle-anchor overflow policy.
- Deciding whether equipment validation is strict.
- Deciding whether backdated activation is allowed.

Do not use this table for:

- Package-specific equipment requirements. Use SIP/PLM catalog data.
- Billing prices or proration formulas. Use BIL configuration and BIL-01.
- Per-customer exceptions. Use operation request data or an approved override flow.
- BPMN process key selection. Use subscription_operation_config from the framework DD.

```
CREATE TABLE subscription_activation_config (
  operator_code                TEXT PRIMARY KEY,
  cycle_anchor_overflow_default TEXT NOT NULL DEFAULT 'PRORATE_PARTIAL_CYCLE',
  customer_self_service_enabled BOOLEAN NOT NULL DEFAULT FALSE,
  sales_field_activation_enabled BOOLEAN NOT NULL DEFAULT TRUE,
  scheduled_activation_enabled  BOOLEAN NOT NULL DEFAULT TRUE,
  equipment_binding_validation_strict BOOLEAN NOT NULL DEFAULT TRUE,
  backdating_allowed            BOOLEAN NOT NULL DEFAULT FALSE,
  backdating_max_days           INTEGER NOT NULL DEFAULT 0,
  backdating_required_role      TEXT NULL,
  updated_at                   TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  updated_by                   TEXT NOT NULL,

  CHECK (cycle_anchor_overflow_default IN (
    'ROUND_TO_28',
    'ROUND_TO_1_NEXT_MONTH',
    'PRORATE_PARTIAL_CYCLE'
  )),
  CHECK (
    (
      backdating_allowed = FALSE
      AND backdating_max_days = 0
      AND backdating_required_role IS NULL
    )
    OR
    (
      backdating_allowed = TRUE
      AND backdating_max_days > 0
      AND backdating_required_role IS NOT NULL
    )
  )
);
```

Column descriptions:

Column	Purpose
operator_code	Operator this activation policy applies to.
cycle_anchor_overflow_default	Default policy when activation day is 29, 30, or 31 and subscription uses CALENDAR_ANCHOR.
customer_self_service_enabled	Allows or blocks activation from customer self-service channels.

Column	Purpose
sales_field_activation_enabled	Allows or blocks activation by field sales/installation agents.
scheduled_activation_enabled	Allows or blocks scheduler-driven activation.
equipment_binding_validation_strict	If true, equipment bindings must be validated against OSR and package requirements before activation proceeds.
backdating_allowed	Whether intendedActivationDate may be before today.
backdating_max_days	Maximum number of days activation may be backdated. Must be zero when backdating is disabled.
backdating_required_role	Role required for backdating, for example SUBSCRIPTION_ADMIN. Must be null when backdating is disabled.
updated_at	Last update timestamp.
updated_by	User/service that last changed the row.

Sample data:

operator_code	cycle_anchor_overflow_default	customer_self_service_enabled	sales_field_activation_enabled	scheduled_activation_enabled	equipment_binding_validation_strict	backdating_allowed	backdating_max_days	backdating_required_role
WIK	PRORATE_PAR TIAL_CYCLE	false	true	true	true	true	7	SUBSCRIPTION_ADMIN
WUG	PRORATE_PAR TIAL_CYCLE	false	true	true	true	false	0	NULL
WTZ	PRORATE_PAR TIAL_CYCLE	false	true	true	true	false	0	NULL
YASSN	PRORATE_PAR TIAL_CYCLE	false	true	true	true	false	0	NULL

Sample meaning:

- WIK allows sales-field activation, scheduled activation, strict equipment validation, and admin-only backdating up to 7 days.
- WUG, WTZ, and YASSN rows are conservative placeholders until their operator-specific flow DDs are written.
- None of the sample operators allows customer self-service activation in v1.0.

7. Business Rules

7.1 Preconditions

Rule	Description
R-ACT-P-1	Subscription must be in CREATED. Otherwise return 409 INVALID_STATE_FOR_ACTIVATION.
R-ACT-P-2	Package must exist and be ACTIVE in SIP-01 at activation time. Otherwise return PACKAGE_RETIRED or PACKAGE_NOT_ACTIVE.
R-ACT-P-3	HomePass must be sellable in RLM-CFG-01. Otherwise return HOMEPASS_NOT_SELLABLE.
R-ACT-P-4	Trigger role must match the trigger and config flags.
R-ACT-P-5	Operation framework must reject duplicate in-flight state-changing operations for the same subscription.
R-ACT-P-6	Idempotent retry with same key and same payload returns the existing operation. Same key with different payload returns 409.

7.2 Cycle Anchor Overflow

Overflow applies only when:

cycle_model = CALENDAR_ANCHOR
activation day is 29, 30, or 31

Policy	Behavior	When to use
ROUND_TO_28	Set cycle_anchor_day = 28.	Simple rule, no special first-cycle logic.

Policy	Behavior	When to use
ROUND_TO_1_NEXT_MONTH	Set <code>cycle_anchor_day = 1</code> for the next month. Billing cycle starts next month.	Operator wants clean monthly billing from day 1.
PRORATE_PARTIAL_CYCLE	Set <code>cycle_anchor_day = 1</code> ; bill partial period from activation date to next day 1.	Operator wants immediate service and fair partial-cycle charge.

Rules:

Rule	Description
R-ACT-OV-1	Caller may provide <code>cycleAnchorOverflowPolicy</code> ; if absent, use <code>subscription_activation_config.cycle_anchor_overflow_default</code> .
R-ACT-OV-2	Resolved <code>cycle_anchor_day</code> must be valid for SUB-LM, which means 1 to 28.
R-ACT-OV-3	For <code>ROLLING_DAYS</code> , no overflow policy applies. Cycle starts from <code>activated_at</code> .
R-ACT-OV-4	BIL-01 computes money for partial cycle. Activation workflow only passes cycle and context facts.

7.3 Equipment Bindings

Equipment binding request item:

```
{
  "equipmentRequirement": "GPON_ONT",
  "equipmentRef": "ont_01HZ_NEW01"
}
```

Rules:

Rule	Description
R-ACT-EB-1	Supplied <code>equipmentRequirement</code> values must match package/service equipment requirements from SIP/PLM.
R-ACT-EB-2	Missing required equipment returns 400 <code>EQUIPMENT_MISSING</code> .
R-ACT-EB-3	Extra equipment not required by the package returns 400 <code>EQUIPMENT_NOT_REQUIRED</code> , unless operator rules explicitly allow it.
R-ACT-EB-4	When strict validation is enabled, <code>equipmentRef</code> must exist in OSR and be available for binding.
R-ACT-EB-5	After fulfillment succeeds, <code>osr-equipment-bind</code> binds equipment to the subscription.
R-ACT-EB-6	Equipment authoritative state is OSR. Activation events may include equipment binding summary for downstream visibility.

7.4 Backdating

Rules:

Rule	Description
R-ACT-BD-1	Backdating is allowed only when <code>backdating_allowed = true</code> .
R-ACT-BD-2	Backdated date must be within <code>backdating_max_days</code> .
R-ACT-BD-3	Backdating requires <code>backdating_required_role</code> .
R-ACT-BD-4	Future activation dates must not be accepted by this endpoint. Use scheduler-driven activation instead.
R-ACT-BD-5	When backdating is accepted, <code>activated_at</code> uses the intended date, but the workflow still runs now.

7.5 Final State Rules

Rule	Description
R-ACT-F-1	Activation success must commit SUB-LM status to <code>ACTIVE</code> .
R-ACT-F-2	Activation failure after pending state must revert subscription to <code>CREATED</code> .
R-ACT-F-3	Activation must emit <code>SubscriptionActivated</code> only after final state commit succeeds.

Rule	Description
R-ACT-F-4	Activation failure must emit SubscriptionActivationFailed after compensation is complete.

8. REST APIs

8.1 Trigger Activation

POST /api/subscriptions/{subscription_id}/activate HTTP/1.1
 Authorization: Bearer <jwt-or-service-token>
 Content-Type: application/json
 Idempotency-Key: act-sub_01HZ_NEW-20260518T100000-sa042

```
{
  "operatorCode": "WIK",
  "activationTrigger": "SALES_FIELD",
  "intendedActivationDate": "2026-05-18",
  "cycleAnchorOverflowPolicy": null,
  "equipmentBindings": [
    {
      "equipmentRequirement": "GPON_ONT",
      "equipmentRef": "ont_01HZ_NEW01"
    },
    {
      "equipmentRequirement": "ATA",
      "equipmentRef": "ata_01HZ_NEW01"
    }
  ],
  "initiatingWorkflowKind": "FUL_02_OBR_01",
  "initiatingWorkflowRef": "obr_01HZ_NEW_CUSTOMER"
}
```

Notes:

- initiatingActorUserId and role should be resolved from auth context, not blindly trusted from body.
- Service callers such as FUL-02 may pass workflow refs, but auth still comes from their service token.

Success:

```
202 Accepted

{
  "operationId": "subop_01HZ_NEW_ACT",
  "operatorCode": "WIK",
  "subscriptionId": "sub_01HZ_NEW",
  "operationKind": "ACTIVATE",
  "bpmnProcessKey": "sub-activate-wik",
  "bpmnProcessInstanceId": "pi_01HZ_NEW_ACT",
  "currentState": "VALIDATING",
  "finalState": null,
  "startedAt": "2026-05-18T10:00:00Z"
}
```

Common errors:

HTTP	Code	Meaning
400	EQUIPMENT_MISSING	Required package equipment was not supplied.
400	EQUIPMENT_NOT_REQUIRED	Request supplied equipment not required by the package.
400	INVALID_CYCLE_ANCHOR_OVERFLOW_POLICY	Unknown overflow policy.
403	ROLE_NOT_AUTHORIZED_FOR_TRIGGER	Actor role cannot use this activation trigger.
409	INVALID_STATE_FOR_ACTIVATION	Subscription is not CREATED.
409	ALREADY_ACTIVE	Subscription is already active.
409	PACKAGE_RETIRED	Package is no longer active.
409	HOMEPASS_NOT_SELLABLE	HomePass cannot be sold/activated.
409	OPERATION_IN_PROGRESS	Another state-changing operation is already running.

8.2 Activation Preview

Preview is read-only. It must not:

- Insert subscription_operation.
- Start BPMN.
- Change subscription state.
- Call fulfillment.
- Bind equipment.

```
POST /api/subscriptions/{subscription_id}/activate/preview HTTP/1.1
Authorization: Bearer <jwt-or-service-token>
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "intendedActivationDate": "2026-05-29",
  "cycleAnchorOverflowPolicy": "PRORATE_PARTIAL_CYCLE"
}
```

Response:

```
{
  "subscriptionId": "sub_01HZ_NEW",
  "resolvedActivatedAt": "2026-05-29",
  "resolvedCycleAnchorDay": 1,
  "cycleAnchorOverflowApplied": "PRORATE_PARTIAL_CYCLE",
  "partialCycleDays": 3,
  "firstCycleStart": "2026-06-01",
  "billing": {
    "lineItems": [
      {
        "kind": "INSTALLATION_FEE",
        "amount": 3500.00,
        "currency": "KES",
        "description": "Installation fee"
      },
      {
        "kind": "PARTIAL_CYCLE_PRORATE",
        "amount": 350.00,
        "currency": "KES",
        "description": "3 days at KES 116.67/day"
      }
    ],
    "totals": {
      "net": 3850.00,
      "currency": "KES"
    }
  }
}
```

9. Worker Usage In This Operation

The worker catalog is defined in DD_SUB-WF-FRAMEWORK-01-WORKERS. This section explains how activation uses those workers.

Important:

The list below is a logical BPMN task list, not a deployment list.

Several logical worker topics can be handled by one worker application. For example, one subscription-state-worker can handle all SUB-LM topics, and one billing-worker can handle all BIL-01 topics. Keep per-topic contracts and metrics, but consolidate runtime deployments where it makes operations simpler.

Worker	Why activation uses it	Key inputs	Key outputs
drools-decide	Validate preconditions, role/trigger, backdating, cycle overflow, equipment policy.	kjarPackage=rules.subscription.activate.<operator>, factsJson, requestedDecisions	decisionsJson
sub-lm-put-pending-status	Move subscription to PENDING_ACTIVATION before external work starts.	subscriptionId, toStatus=PENDING_ACTIVATION, currentTransitionType=ACTIVE	priorStatusSnapshot, lmEventId
sub-lm-compute-cycle-window	Compute cycle dates for billing and event payloads.	subscriptionId, referenceDate=activatedAt	cycleWindowJson

Worker	Why activation uses it	Key inputs	Key outputs
bil01-charge-pre-authorize	Create pay-first activation/installation/first-cycle charge when operator BPMN requires payment before service.	subscriptionId, chargeCode=ACTIVATION_INTENT, chargeContext	billableEventId, chargeCreatedAt
bil01-confirm-charge	Confirm the pending charge after payment is received.	billableEventId	confirmedAt, chargeStatus
bil01-revert-pending-charge	Cancel pending charge during activation compensation.	billableEventId, cancelReason	cancelledAt
ful-call-activate	Ask FUL-03 to provision the service.	subscriptionId, activationCorrelationId, equipmentBindings	activationFulfillmentId, activationTerminalState
ful-call-claim-homepass	Ask FUL-07 to claim HomePass capacity.	subscriptionId, homepassId, claimReason	homepassClaimId, incrementedActiveTerminationPoints
osr-equipment-bind	Bind ONT/STB/ATA/etc. equipment to the subscription.	subscriptionId, equipmentBindings, bindCorrelationId	boundCount, boundInstanceRefs
osr-equipment-unbind	Undo equipment binding in compensation.	subscriptionId, equipmentBindings, unbindCorrelationId	unboundCount
sub-lm-put-master-fields	Persist activation master fields in SUB-LM.	subscriptionId, fieldsJson	lmEventId
sub-lm-commit-state	Commit subscription to ACTIVE.	subscriptionId, toStatus=ACTIVE, expectedFromStatus=PENDING_ACTIVATION	committedAt, lmEventId
sub-lm-revert-to-prior	Revert to CREATED if activation fails after pending state.	subscriptionId, priorStatus=CREATED, failureReasonCode	revertedAt, lmEventId
kafka-emit	Write SubscriptionActivated or SubscriptionActivationFailed to outbox.	topic, key=subscriptionId, payloadJson	outboxId
notification-send	Send welcome/failure notification. Does not fail activation if notification fails.	recipientKind, recipientRef, notificationEventKind, eventContext	notificationId, dispatched
subscription-operation-finalize	Mark operation COMPLETED or FAILED.	operationId, finalState, failure fields when needed	finalizedAt

Operator note:

- WIK uses pay-first workers.
- An operator without pay-first activation can skip bil01-charge-pre-authorize, payment-confirmed, and bil01-confirm-charge.
- If equipment is not required for a package, the BPMN can skip OSR bind/unbind workers.

10. Kafka Events

10.1 SubscriptionActivated

Topic:

sophix.subscription.subscription.activated

When emitted:

- After FUL-03 activation succeeds.
- After required HomePass/equipment steps are complete or accepted by their owning modules.
- After SUB-LM commits status to ACTIVE.

Sample:

```
{
  "eventType": "SubscriptionActivated",
  "aggregateId": "sub_01HZ_NEW",
  "timestamp": "2026-05-18T10:00:18Z",
  "payload": {
    "operatorCode": "WIK",
    "subscriptionId": "sub_01HZ_NEW",
```

```

"customerId": "cust_01HZ_NEW",
"homepassId": "hp_01HZ_NRB",
"operationId": "subop_01HZ_NEW_ACT",
"activationTrigger": "SALES_FIELD",
"packageRef": "pkg_INET_VOIP_RES",
"packageVersionId": "pkgver_003",
"activatedAt": "2026-05-18T10:00:18Z",
"cycleModel": "CALENDAR_ANCHOR",
"resolvedCycleAnchorDay": 1,
"cycleAnchorOverflowApplied": "PRORATE_PARTIAL_CYCLE",
"partialCycleDays": 13,
"equipmentBindings": [
  {
    "equipmentRequirement": "GPON_ONT",
    "equipmentRef": "ont_01HZ_NEW01"
  },
  {
    "equipmentRequirement": "ATA",
    "equipmentRef": "ata_01HZ_NEW01"
  }
],
"billingMode": "PREPAID",
"contractStartedAt": "2026-05-18T10:00:18Z",
"contractMinimumTermMonths": 12,
"contractEndAt": "2027-05-18T10:00:18Z",
"initiatingActorUserId": "sales-agent-wik-042",
"initiatingActorRole": "SALES_AGENT"
}
}

```

Consumers:

Consumer	Use
NOT-01	Welcome notification.
BIL-02	Begin invoicing cycle.
BIL-05	Open prepaid wallet when billing mode is prepaid.
Reporting/analytics	New activation reporting.

10.2 SubscriptionActivationFailed

Topic:

sophix.subscription.subscription.activationfailed

When emitted:

- Activation cannot complete.
- Compensation has run.
- Subscription has been reverted to CREATED when it had already entered PENDING_ACTIVATION.

Sample:

```

{
  "eventType": "SubscriptionActivationFailed",
  "aggregateId": "sub_01HZ_NEW",
  "timestamp": "2026-05-18T10:00:45Z",
  "payload": {
    "operatorCode": "WIK",
    "subscriptionId": "sub_01HZ_NEW",
    "operationId": "subop_01HZ_NEW_ACT_FAIL",
    "activationTrigger": "SALES_FIELD",
    "rejectionReasonCode": "FULFILLMENT_FAILED",
    "rejectionReasonDetail": "FUL-03 returned terminal failure during GPON activation.",
    "subscriptionRevertedTo": "CREATED",
    "initiatingActorUserId": "sales-agent-wik-042"
  }
}

```

Consumers:

Consumer	Use
NOT-01	Notify agent/customer where policy allows.
ICN-01	Operational alert.
Reporting/analytics	Failed activation reporting.

11. Drools Package

Package name pattern:

```
rules.subscription.activate.<operator>
```

Examples:

```
rules.subscription.activate.wik
rules.subscription.activate.wug
rules.subscription.activate.wtz
rules.subscription.activate.yassn
```

Activation rules should return decisions, not perform side effects.

Expected requested decisions:

Decision	Purpose	Example output
validate-preconditions	Confirms subscription/package/HomePass/role are valid.	{"eligible":true}
validate-backdating	Confirms intended activation date is allowed.	{"backdatingValid":true,"backdatingApplied":false}
resolve-cycle-anchor-overflow	Resolves anchor day and partial-cycle facts.	{"resolvedCycleAnchorDay":1,"partialCycleDays":3}
validate-equipment-bindings	Confirms equipment request matches operator/package policy.	{"equipmentValid":true}

Sample facts:

```
{
  "operatorCode": "WIK",
  "subscriptionStatus": "CREATED",
  "packageStatus": "ACTIVE",
  "homepassSellable": true,
  "activationTrigger": "SALES_FIELD",
  "initiatingActorRole": "SALES_AGENT",
  "intendedActivationDate": "2026-05-18",
  "cycleModel": "CALENDAR_ANCHOR",
  "equipmentBindings": [
    {
      "equipmentRequirement": "GPON_ONT",
      "equipmentRef": "ont_01HZ_NEW01"
    }
  ],
  "activationConfig": {
    "salesFieldActivationEnabled": true,
    "equipmentBindingValidationStrict": true
  }
}
```

Sample decisions:

```
{
  "eligible": true,
  "backdatingApplied": false,
  "backdatingValid": true,
  "resolvedCycleAnchorDay": 18,
  "partialCycleDays": 0,
  "overflowApplied": null,
  "equipmentValid": true
}
```

12. Auth

Activation endpoint must use FOUNDATION_AUTH.

Rules:

- Resolve actor user and role from JWT/service token.
- Do not trust actor role from request body.
- Service-to-service calls must use service accounts.
- CUSTOMER_SELF_SERVICE trigger requires both role and config flag.
- SALES_FIELD trigger requires both role and config flag.

- Backdating requires the configured role.

Additional group required by activation:

Group	Purpose
SALES_AGENT	Allows field sales/installation activation when enabled by config.

If SALES_AGENT is not yet in FOUNDATION_AUTH, add it as a subscription workflow group before production activation rollout.

13. Error Handling

Error code	HTTP/status	Meaning
INVALID_STATE_FOR_ACTIVATION	409	Subscription is not CREATED.
ALREADY_ACTIVE	409	Subscription is already ACTIVE.
SUBSCRIPTION_TERMINATED	409	Subscription is terminal and cannot be activated.
PACKAGE_RETIRED	409	Package is no longer active.
HOMEPASS_NOT_SELLABLE	409	HomePass cannot be sold/activated.
ROLE_NOT_AUTHORIZED_FOR_TRIGGER	403	Actor role cannot use requested trigger.
CUSTOMER_SELF_SERVICE_ACTIVATION_DISABLED	403	Self-service trigger disabled for operator.
SALES_FIELD_ACTIVATION_DISABLED	403	Sales-field trigger disabled for operator.
BACKDATING_NOT_ALLOWED	403 or 400	Backdating disabled or role missing.
BACKDATING_RANGE_EXCEEDED	400	Intended date is too far in the past.
FUTURE_ACTIVATION_DATE_NOT_ALLOWED	400	Use scheduler instead.
EQUIPMENT_MISSING	400	Required equipment not supplied.
EQUIPMENT_NOT_REQUIRED	400	Extra equipment supplied.
EQUIPMENT_BINDING_INVALID	400 or 409	Equipment does not exist or cannot be bound.
ACTIVATION_PAYMENT_TIMEOUT	Final failure	Payment was not confirmed before BPMN timer.
FULFILLMENT_FAILED	Final failure	FUL-03 returned terminal failure.

14. Implementation Notes

14.1 Trigger Endpoint Sequence

Use this sequence:

1. Authenticate caller.
2. Resolve actor user ID, role, and operator scope from auth context.
3. Validate Idempotency-Key.
4. Validate request shape.
5. Load subscription from SUB-LM/read model.
6. Load subscription_activation_config.
7. Load framework subscription_operation_config for (operator_code, ACTIVATE).
8. Validate trigger role and config flags.
9. Validate subscription is CREATED.
10. Validate package and HomePass references.
11. Validate equipment request shape.
12. Call framework start operation contract with operationKind=ACTIVATE.
13. Framework creates subscription_operation and starts BPMN.
14. Return 202 Accepted.

Keep trigger endpoint logic thin. Detailed orchestration belongs in BPMN.

14.2 Idempotency

Activation trigger must use framework idempotency:

```
(operator_code, idempotency_key)
```

Same key and same request:

```
return existing operation
```

Same key and different request:

```
409 IDEMPOTENCY_KEY_REUSED_WITH_DIFFERENT_REQUEST
```

Recommended idempotency key from FUL-02:

```
activation:<subscriptionId>:<onboardingRef>
```

14.3 Activation Date Resolution

Compute these values before billing and final commit:

Value	Meaning
resolvedActivatedAt	Date/time that business treats as activation time.
resolvedCycleAnchorDay	SUB-LM-safe anchor day, 1 to 28.
cycleAnchorOverflowApplied	Null or selected overflow policy.
partialCycleDays	Days to pass to billing for partial-cycle calculation.

Do not let SUB-LM accept invalid anchor days. Activation must resolve days 29-31 before calling `sub-lm-put-master-fields`.

14.4 Equipment Validation

Validation has two layers:

1. Shape validation in API/BPMN: array exists, fields are present, no duplicate `equipmentRequirement`.
2. Business validation through catalog/OSR: required equipment matches package, equipment instances exist and can be bound.

Do not bind equipment before network activation succeeds unless the operator BPMN explicitly requires early reservation.

14.5 Compensation

Minimum compensation behavior:

Failure point	Compensation
Validation fails before pending state	Finalize failed; no subscription revert needed.
Pending state set, billing fails	<code>sub-lm-revert-to-prior</code> , finalize failed.
Payment timeout	<code>bil01-revert-pending-charge</code> , <code>sub-lm-revert-to-prior</code> , finalize failed.
Fulfillment fails	<code>bil01-revert-pending-charge</code> , <code>sub-lm-revert-to-prior</code> , finalize failed.
Equipment bind fails after fulfillment	Reverse fulfillment if available, revert charge, revert subscription, finalize failed.
SUB-LM commit fails after fulfillment	Route to manual recovery; do not silently mark completed.

14.6 Event Timing

Emit `SubscriptionActivated` only after:

1. SUB-LM commit to ACTIVE succeeded.
2. Required billing confirmation succeeded, if `pay-first` is used.
3. Required fulfillment succeeded.
4. Required equipment binding succeeded or was explicitly skipped by policy.

Do not emit SubscriptionActivated from the trigger endpoint.

15. Infrastructure Recommendations

Activation does not require separate infrastructure from SUB-WF.

Deploy these artifacts with the subscription workflow service:

Artifact	Where it runs
Activation endpoint/controller	SUB-WF API pods.
Activation config migration	Application DB migration pipeline.
Activation BPMN files	Camunda deployment pipeline.
Activation Drools packages	Drools/KIE deployment pipeline.
Activation worker usage	Worker pods from the WORKERS DD.

Medium deployment guidance:

- Use the SUB-WF API and worker sizing already defined in the framework and worker DDs.
- Ensure at least two worker replicas can process activation topics.
- Ensure activation worker topics have dashboards for backlog, failure rate, and p95 duration.
- Ensure FUL-03 and BIL-01 latency stays below worker lock-duration safety margins.

AWS guidance:

- No separate ECS/EKS service is required just for activation.
- Deploy activation BPMN and rules as versioned artifacts.
- Use CloudWatch alarms for activation failure rate, payment timeout count, fulfillment failure count, and stuck PENDING_ACTIVATION subscriptions.

16. Cross-DD Dependencies

DD/module	Relationship
SUB-WF-FRAMEWORK-01	Creates operation row, enforces idempotency/concurrency, starts BPMN.
DD_SUB-WF-FRAMEWORK-01-WORKERS	Defines workers used by activation.
DD_SUB-WF-ACTIVATE-01-FLOW-WIK	WIK reference BPMN for this operation.
SUB-LM-01	Owns subscription status and master fields.
SIP-01	Package active/version validation.
PLM-CFG-01	Service/equipment requirement catalog.
RLM-CFG-01	HomePass sellability validation.
BIL-01	Activation billing intent and pay-first charge.
BIL-02	Consumes activation event for invoice cycle.
BIL-05	Opens prepaid wallet after activation event.
FUL-02	Upstream onboarding caller.
FUL-03	Network activation fulfillment.
FUL-07	HomePass claim.
OSR-INSTANCE-01	Equipment binding.
NOT-01	Welcome/failure notifications.
FOUNDATION_AUTH	Roles and service accounts.
FOUNDATION_DROOLS	Activation rule execution.

17. Build Order

The detailed implementation procedure is in DD_SUB-WF-ACTIVATE-01-DEV_IMPL_GUIDE-v1.0.md.

Recommended build order:

1. Add activation config migration and seed rows.
2. Add framework operation config for ACTIVATE.
3. Implement DTOs, controller, and activation service.
4. Implement preview endpoint.
5. Implement Drools fact and decision mapping.
6. Deploy the first BPMN, normally sub-activate-wik.
7. Wire BPMN service tasks to worker topics.
8. Add event builders.
9. Add compensation paths.
10. Run unit, service, BPMN, and manual smoke tests from the dev guide.

18. Test Checklist

- [] Activation rejects non-CREATED subscriptions.
- [] Activation rejects retired/inactive package.
- [] Activation rejects non-sellable HomePass.
- [] Trigger role gating works for every trigger.
- [] Disabled config flag blocks corresponding trigger.
- [] Idempotent retry returns existing operation.
- [] Same idempotency key with different payload returns 409.
- [] Overflow day 29/30/31 resolves correctly for each policy.
- [] ROLLING_DAYS ignores overflow policy.
- [] Backdating disabled rejects past date.
- [] Backdating enabled requires configured role.
- [] Equipment missing returns EQUIPMENT_MISSING.
- [] Equipment invalid returns EQUIPMENT_BINDING_INVALID.
- [] Preview does not create operation or mutate state.
- [] Success commits subscription to ACTIVE.
- [] Success emits SubscriptionActivated only after commit.
- [] Failure after pending state reverts to CREATED.
- [] Failure emits SubscriptionActivationFailed after compensation.
- [] Notification failure does not fail activation.

19. Developer Checklist

- [] Do not activate by directly updating subscription.status_code.
- [] Use SUB-WF framework start contract.
- [] Use shared workers by topic name.
- [] Keep billing calculation in BIL.
- [] Keep network provisioning in FUL-03.
- [] Keep equipment state in OSR.
- [] Keep large payloads out of Camunda variables.
- [] Emit domain events through kafka-emit/outbox.
- [] Keep operator-specific variation in BPMN, Drools, and config.